

Consumer Electronics Market Intelligence

Complete Insight Report | Laptops & Smartphones

Purpose: Translate an interactive Excel market-analysis dashboard into business-oriented findings, questions, implications, and recommendations.

Key Performance Indicators (KPIs)

Products	Brands	Avg. Price	Avg. RAM	Avg. Storage	Avg. Rating
15,991	33	45,565.17	7.83	264.2	63.9

Report note: The dashboard contains multiple filter states. The headline market baseline shown in the overview is 15,991 products; the second screenshot demonstrates a filtered state where the KPI values change. Therefore, filter-specific KPI values should not be interpreted as the overall market baseline.

Prepared from the Excel dashboard and dashboard screenshots provided for the project.

Updated dashboard date shown: 8 August 2026

1. Executive Summary

The Consumer Electronics Market Intelligence project analyzes laptops and smartphones through an interactive Microsoft Excel workflow. The project is designed as more than a visual dashboard: Power Query is used for data preparation, Pivot Tables provide the analytical layer, Pivot Table connections and slicers drive interactivity, and a dedicated insight report converts the results into business-oriented conclusions.

At the overall dashboard level, the analysis covers 15,991 products across 33 brands. The baseline dashboard reports an average price of 45,565.17, average RAM of 7.83, average storage of 264.2, and an average rating of 63.9.

The strongest high-level signal is that the market is broad and specification-driven. Smartphones form the larger product category in the displayed category comparison, while Android dominates the operating-system distribution. The price mix is concentrated toward mid-range and premium products, and Performance and Standard configurations account for large product volumes.

Overall business takeaway: the dashboard can support market assortment analysis, brand benchmarking, price positioning, and product-specification decisions. Its greatest value comes from combining interactive filtering with business questions and an accompanying interpretation report.

2. Data & Analytical Methodology

2.1 End-to-End Workflow

The project follows a repeatable Excel analytics pipeline:

Stage	How it is used	Business value
1. Source data	Product-level consumer electronics records	Creates the market-level analytical base
2. Power Query	Clean, transform, and prepare data	Improves consistency and makes refreshes easier
3. Pivot Tables	Aggregate products by brand, category, price, OS, RAM, storage, processor, etc.	Provides ready summaries
4. Pivot connections	Connect analytical Pivot Tables to dashboard controls	Keeps dashboard elements synchronized
5. Slicers / filters	Filter by category, brand, price segment, OS, RAM category, etc.	Enables interactive investigation
6. Dashboard	KPIs + charts + key insight panels	Provides fast market-level understanding
7. Insight report	Interpret patterns and implications	Moves from visualization to business reasoning

2.2 Analytical Dimensions

The dashboard combines market, commercial, and technical dimensions: product category, brand, price segment, operating system, RAM category, storage category, processor brand, and rating. This allows the same dataset to be viewed from both a market-positioning and product-specification perspective.

3. Market Baseline & KPI Interpretation

Metric	Baseline value	Interpretation
Total Products	15,991	Large product assortment across laptops and smartphones.
Brands	33	Provides a broad basis for brand presence and positioning comparisons.
Average Price	45,565.17	Useful baseline for comparing product segments and brands.
Average RAM	7.83	Indicates the overall specification level of the analyzed portfolio.
Average Storage	264.2	Provides a market-level storage benchmark.
Average Rating	63.9	Can be used as a broad quality/customer-perception reference.

The KPI cards should be treated as dynamic measures rather than fixed facts. When a slicer is applied, the same measures recalculate for the selected subset. This is a key strength of the dashboard because it supports both overall-market and segment-level analysis.

4. Market Structure Insights

4.1 Category Mix

The Product by Category chart shows a strong concentration of the analyzed product portfolio in smartphones, while laptops form a much smaller portion of the displayed total. This suggests that smartphone assortment is the dominant component of the current dataset.

Business implication: A business using this dataset should avoid treating the combined market average as representative of both categories equally. Category-level KPIs are important because the product mix is uneven.

4.2 Operating System

Android is visually dominant in the operating-system distribution. iOS represents a much smaller share in the displayed chart, while the laptop-focused operating systems appear as smaller segments.

Business implication: Android-oriented products represent the largest addressable operating system group in the current dataset. However, operating-system share should be evaluated separately for smartphones and laptops to avoid mixing fundamentally different device ecosystems.

4.3 Price Segmentation

The price-segment chart shows the largest product volumes in the Premium and Mid-Range groups, followed by Budget, with Flagship representing a smaller volume in the displayed overall view.

Business implication: The market appears to have substantial assortment depth beyond entry-level products. Pricing strategy should therefore consider both volume and positioning rather than assuming that lower-priced products dominate simply because they target a broader customer base.

5. Brand & Competitive Insights

The Top Brands by Product Count chart provides a view of assortment breadth. The displayed rankings change when the Brand or Category slicers are used, demonstrating that brand presence is highly dependent on the selected market slice.

The dashboard also includes an Average Price by Brand (Top 10) analysis. This is important because product count and average price measure two different competitive strategies: a brand may have broad

assortment without having the highest average price, while another may have fewer products but stronger premium positioning.

Key competitive questions:

- Which brands combine high product volume with high average pricing?
- Which brands are concentrated in Budget, Mid-Range, Premium, or Flagship segments?
- Does a brand's product mix differ substantially between laptops and smartphones?
- Which brands appear specification-focused rather than price-focused?

6. Product Specification Insights

6.1 RAM

The RAM chart shows the highest displayed product count for the Performance category, followed by Entry Level, Standard, and Flagship. This indicates that the dataset contains substantial assortment around performance-oriented configurations.

6.2 Storage

Standard and Ultra storage categories show the highest displayed product volumes, while Premium is the smallest of the four shown groups. This provides a useful basis for understanding how storage capacity is distributed across the market.

6.3 Processor Brand

The processor chart shows Apple and Other as major groups, with MediaTek also represented strongly. Because processor ecosystems differ across device categories, the most useful next step is to compare processor mix separately for laptops and smartphones.

Business implication: Specification-level analysis can help identify the configurations with the largest product supply and can support product benchmarking, assortment planning, and competitive positioning.

7. Business Questions & Decision Framework

The project is intentionally structured around business questions. The dashboard is the exploration layer; the following questions define how an analyst could use it for decision-making.

Business area	Question	Potential decision
Market	Which category dominates the current product portfolio?	Allocate analysis and resources according to market weight.
Brand	Which brands have the broadest product presence?	Benchmark assortment breadth and competitive intensity.
Pricing	Which brands and segments command higher average price?	Formulate premium positioning and pricing strategy.
Specifications	Which RAM and storage segments have the greatest product volume?	Identify common configurations and market supply.
OS	Which operating systems dominate each category?	Understand ecosystem concentration and competitive exposure.
Processors	Which processor groups have the strongest presence?	Support component-level competitive analysis.
Category comparison	How different are laptops and smartphones?	Avoid misleading combined-market conclusions.
Filters	How do the KPIs change for a selected brand/category?	Perform focused segment-level analysis.

8. Interactive Filter Analysis

The two dashboard screenshots demonstrate why filter-driven analysis is important. The first view represents the broader market overview, while the second shows a different selection of Category, Brand, Price Segment, Operating System, and RAM Category filters. The KPI values and chart distributions change accordingly.

Analytical lesson: a dashboard should not be judged only by its default view. Its real value is its ability to answer follow-up questions quickly by changing the selected market slice.

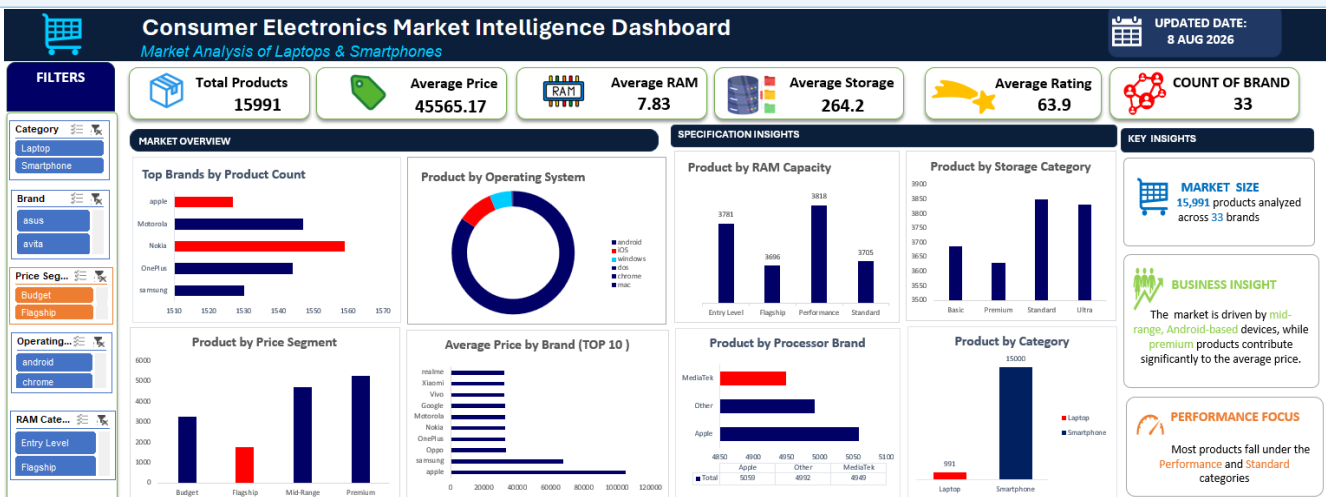


Figure 1. Overall dashboard view.

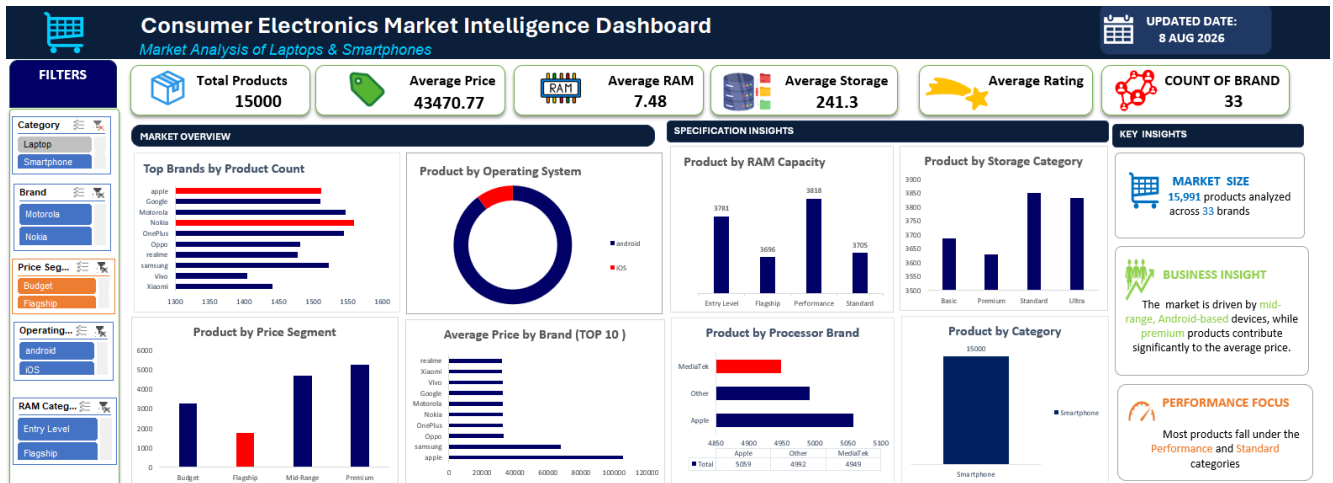


Figure 2. Example filtered dashboard view.

9. Business Recommendations

1. Use category-specific benchmarks

Because smartphones dominate the displayed product mix, overall averages should always be paired with category-level averages. This will make pricing and specification comparisons more meaningful.

2. Segment brands by assortment and pricing

Combine product count with average price. This can distinguish broad-market brands from brands that rely on premium positioning.

3. Build specification benchmarks

RAM, storage, processor, and rating should be analyzed together with price to identify price-performance opportunities.

4. Use the slicers as an investigation tool

For practical decision-making, analysts can begin with the overall market and then drill into a category, brand, or price segment. This makes the dashboard suitable for exploratory business analysis.

5. Maintain a separate insight report

The dashboard should remain concise and visual, while the report should document the reasoning, business implications, assumptions, and recommendations. This separation improves communication with non-technical stakeholders.

10. Future Enhancement Roadmap

The long-term objective is to expand the current Laptop + Smartphone analysis into a complete Consumer Electronics Market Intelligence system.

Phase	Expansion	Expected value
Current	Laptops + Smartphones	Core market intelligence foundation
Next	Add Tablets	Extend cross-category comparison
Next	Add Smartwatches	Cover wearable electronics
Advanced	Cross-category benchmarking	Compare pricing and specifications across devices
Advanced	Trend / time analysis	Track changes in market structure over time
Advanced	Automated refresh	Reduce manual reporting effort
Advanced	Advanced KPIs	Add price-performance and brand-positioning metrics

11. Limitations & Interpretation Notes

This report is based on the values and visual patterns available in the provided dashboard screenshots. It does not independently recalculate the underlying dataset. Therefore, detailed statistical conclusions such as market share percentages, correlations, or category-specific averages should be calculated from the source workbook before being presented as exact figures.

The dashboard screenshots also show different filter states. For example, one view reports 15,991 total products while another filtered view reports 15,000. This is expected behavior for a connected interactive

dashboard and demonstrates why filter context must always be stated when interpreting a KPI.

Recommended reporting practice: always label findings as either overall market or filtered segment and record the active slicer selections when sharing a specific insight.

12. Conclusion

The Consumer Electronics Market Intelligence project demonstrates an end-to-end approach to Excel-based business analytics. The combination of Power Query, Pivot Tables, Pivot Table connections, slicers, dynamic KPIs, visual analysis, and a dedicated insight report turns raw product data into a reusable decision-support workflow.

The current analysis identifies smartphones as the dominant displayed category, Android as the dominant operating-system group, strong product concentration in Mid-Range/Premium pricing, and substantial assortment in Performance and Standard specification categories. More importantly, the interactive design allows these observations to be tested across different brands, categories, price segments, operating systems, and RAM groups.

Final takeaway: the strongest next step is not simply adding more charts. It is expanding the dataset to Tablets and Smartwatches, adding stronger cross-category benchmarks, and continuing to connect business questions with measurable KPIs and documented insights.